

Solutions HW07

1/

1. (a) P_1 is pos. semi def because

$$\lambda_{\min}(P_1) = 0.$$

$$P_1 = N^T N, \text{ where } N = \begin{bmatrix} 1 & 3 \end{bmatrix}$$

$$N = \begin{bmatrix} 1 & 3 \end{bmatrix}.$$

Also OK to say $N = \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix}$

(b) P_2 is pos. def. because $\lambda_{\min}(P_2) = 0.1856$.

$$P_2 = N^T N \text{ where } N = \begin{bmatrix} -0.3767 & 0.1409 & 0.1543 \\ -0.0255 & -0.7991 & 0.6671 \\ 3.2401 & 3.6886 & 4.5417 \end{bmatrix}$$

other answers are possible, such as

$$N = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

(c) P_3 is not pos. semi. def. because

$$\lambda_{\min}(P_3) = -2,9165 < 0.$$

$$(a) \quad P_1 > 0 \iff 1 > 0 \text{ and } 8 - 3(1)3 > 0$$

$$\text{But } 8 - 9 = -1 < 0 \quad \therefore \text{Not Pos. Def.}$$

$$(b) \quad P_2 > 0 \iff A = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} > 0 \text{ and}$$

$$10 - [6 \ 7] \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 7 \end{bmatrix} > 0.$$

$A > 0$ is obvious.

$$10 - [6, 7] \begin{bmatrix} 1 & 0 \\ 0 & 1/4 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} = -38,25$$

$\therefore$ Not Pos Def.

$$(c) \quad P_3 > 0 \iff \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} > 0 \text{ and}$$

$$a - [6 \ 7] \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 7 \end{bmatrix} > 0$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} > 0 \quad \text{because } 1 > 0 \text{ and } 5 - 4 > 0$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}^{-1} = \frac{1}{1} \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$$

$$a - [6 \ 7] \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} > 0 \quad \text{if, and only if}$$

$$a - [6 \ 7] \begin{bmatrix} 16 \\ -5 \end{bmatrix} = a - 61 > 0$$

$$\Rightarrow \boxed{a > 61}$$

3. For $\langle x, y \rangle = x^T Q y$, the minimum norm solution is

$$\hat{x} = Q^{-1} A^T (A Q^{-1} A^T)^{-1} b$$

a) $Q = I$, and we have

$$\hat{x} = \begin{bmatrix} -0.0952 \\ 0.0476 \\ 0.4762 \end{bmatrix}$$

b) $Q \neq I$, and we have

$$\hat{x} = \begin{bmatrix} -0.6497 \\ 0.3248 \\ 0.3376 \end{bmatrix}$$

4.

(a) }
(b) }
(c) }

Typeset data in Prob4.txt

5.

(a) }
(b) }
(c) }
(d) }

See Prob5.txt

b(a) From the handout, we

identify $X_1 = \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$, $X_2 = Z$,

$$\Sigma_{11} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \quad \Sigma_{12} = 0, \quad \Sigma_{21} = \Sigma_{12}^T,$$

$$\Sigma_{22} = 1$$

Hence, μ_{12}

$$\mu_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad \mu_2 = [1]$$

$$\therefore \mu_{12} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (Z - \mu_2)$$

$$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} (Z - 1)$$

$$= \begin{bmatrix} -3/2 + Z/2 \\ Z - 1 \end{bmatrix}$$

$$\Sigma_{112} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} = \begin{bmatrix} 3/2 & 1 \\ 1 & 2 \end{bmatrix}$$

(b) We have

$$\mu = \mu_{1|2} = \begin{bmatrix} -\frac{3}{2} + \frac{z}{2} \\ z-1 \end{bmatrix}$$

$$\Sigma = \Sigma_{1|2} = \begin{bmatrix} \frac{3}{2} & 1 \\ 1 & 2 \end{bmatrix}$$

We apply the handout to compute

$$\mu_{1|2} = \left(-\frac{3}{2} + \frac{z}{2}\right) + (1)\left(\frac{1}{2}\right)(y - (z-1)) = -1 + \frac{y}{2}$$

$$\Sigma_{1|2} = \frac{3}{2} - (1)\left(\frac{1}{2}\right)(1) = \frac{3}{2} - \frac{1}{2} = 1$$

◦◦ The random variable $X|_{Z=z}$ conditioned on $Y|_{Z=z}=y$ has a normal distribution with

$$\text{Mean} = -1 + \frac{y}{2}$$

$$\sigma^2 = 1$$

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(c) We go back to the beginning and identity

$$X_1 = X, \quad X_2 = \begin{bmatrix} y \\ z \end{bmatrix}, \quad x_2 = \begin{bmatrix} y \\ z \end{bmatrix}$$

$$\Sigma_{11} = 2, \quad \Sigma_{12} = [2 \ 0]$$

$$\Sigma_{21} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \Sigma_{22} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\mu_1 = -1 \quad \text{and} \quad \mu_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\sigma_0 \mu_{1|2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} \left(\begin{bmatrix} y \\ z \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \quad \mu_2$$

$$= -1 + \begin{bmatrix} \frac{1}{2} & 0 \end{bmatrix} \begin{pmatrix} y \\ z - 1 \end{pmatrix}$$

$$= -1 + \frac{y}{2}$$

$$\sigma^2 = 2 - [2 \ 0] \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 1$$

(d) They are the same!